

# . Intro to Stata

A hands-on session

University of Groningen

Ade Febriady

Friday 11 September 2026 · 11:00–12:45

## . An email you will get



**your supervisor**

Subject: five numbers by Thursday

Tue 13 Apr 2027, 09:12

Hi,

I am writing the wage section and need five things from the survey by Thursday: how many people we have, what a typical worker earns, who finished secondary school, the return to schooling, and whether it differs for people who grew up poor.

Send me the numbers and how you got them. Thanks!

**Reply**

by 12:45 you can answer it

## . Why Stata

journals

**73%**

of the code packages in the  
AEA archive use Stata<sup>1</sup>

surveys

**.dta**

LSMS and DHS are available  
as Stata files<sup>2</sup>

jobs

**.do**

employers, e.g. the World  
Bank, set Stata tests and ask  
to see your do-file<sup>3</sup>

the habit, a script and a log, moves to R or Python unchanged

<sup>1</sup> Vilhuber, Turitto and Welch (2020), *AEA Papers and Proceedings* 110: 764–775, corrigendum of 4 Aug 2020.

<sup>2</sup> DHS Program, File Formats; World Bank Microdata Library. <sup>3</sup> Personal experience, World Bank consultancy recruitment.

## . You talk to Stata in commands

three ways to give one, and today is the third

|                |                                                                                                                     |
|----------------|---------------------------------------------------------------------------------------------------------------------|
| menus          | click through the drop-down menus. Stata prints the command each click ran, so you can copy it                      |
| Command window | type one command, press Enter, read the answer                                                                      |
| do-file        | write the commands in a text file, run the file. Your work is saved, repeatable, and can be checked by someone else |

- **The university gives you Stata two ways**

`uwp.rug.nl`

the university's virtual workplace.  
sign in with your student account

IRIS

download and install it  
on your own machine, for later

# Stata is four windows and one editor

The screenshot shows the Stata MP 19.5 interface with the following components highlighted:

- Do-file Editor:** The top window showing the command history and results.
- history:** The left window showing the command history.
- results:** The middle window showing the results of the commands.
- variables:** The right window showing the list of variables.
- command:** The bottom window showing the command line.
- properties: ignore:** The bottom right window showing the properties of the current frame.

The Do-file Editor window contains the following text:

```
. cd "/Users/adefebrady/Downloads/intro_to_stata"
/Users/adefebrady/Downloads/intro_to_stata

. use "wage_survey.dta", clear
(SIMULATED wage survey - Intro to Stata session (NOT real IFLS data))

. describe

Contains data from wage_survey.dta
Observations:      8,000      SIMULATED wage survey - Intro to Stata session (NOT real IFLS data)
Variables:         10        4 Sep 2026 15:35
                        (_dta has notes)
```

The results window displays the following table:

| Variable name | Storage type | Display format | Value label | Variable label                                            |
|---------------|--------------|----------------|-------------|-----------------------------------------------------------|
| id            | int          | %12.0g         |             | Respondent identifier                                     |
| hwage         | double       | %10.0g         |             | Hourly wage (Rupiah)                                      |
| schooling     | byte         | %8.0g          |             | Years of schooling completed                              |
| cpoor         | byte         | %8.0g          | yesno       | Grew up poor (childhood poverty indicator)                |
| female        | byte         | %8.0g          | yesno       | Female                                                    |
| urban         | byte         | %8.0g          | yesno       | Lives in urban area                                       |
| age           | int          | %8.0g          |             | Age at survey                                             |
| birthyr       | int          | %8.0g          |             | Year of birth (cohort)                                    |
| province      | byte         | %10.0g         | provlbl     | Province of residence                                     |
| ability       | double       | %10.0g         |             | Latent ability (UNOBSERVABLE in real data; Module C only) |

The variables window shows the following list of variables:

| Name      | Label        |
|-----------|--------------|
| id        | Respond...   |
| hwage     | Hourly w...  |
| schooling | Years of...  |
| cpoor     | Grew up...   |
| female    | Female       |
| urban     | Lives in...  |
| age       | Age at s...  |
| birthyr   | Year of b... |
| province  | Province...  |
| ability   | Latent a...  |

The command window shows the following command:

```
Command
```

The properties window shows the following properties:

| Properties |
|------------|
| Variables  |
| Name       |
| Label      |

the windows are Stata talking to you; the editor, one click away, is you talking back.  
On the lab's Windows machines the boxes sit in a different order; the four are the same

## . Click once, then copy what Stata typed

`File > Open` opens the data by clicking. Look at the Results window: Stata prints the command it just ran

`use "wage_survey.dta", clear` that printed line. Copy it into your do-file, and it is yours forever

`Statistics > Summaries > Summary statistics` the same trick for any command you do not know yet

the menus are a way to find a command; the do-file is where you keep it

- You write the do-file; Stata writes the log**



keep both; next week's you can re-read today

## . The anatomy of a do-file

`* a note for people` Stata skips everything after a star

`summarize // a note at the end of a line` two slashes do the same, after a command

`/* a longer note, over several lines */` for a paragraph

`**# 3. Look at the data` a bookmark. The editor lists them, so a long file has a table of contents

`cd "..."` the first real command: where your folder is

`log using "session.log", replace` the second: keep a record of everything from here

notes first, then the two lines every do-file starts with, then the work

## . Four kinds of file

`.dta` data, in Stata's own format. `use` opens one, `save` writes one

`.do` a do-file: your commands, in order, with your notes. The thing you write today

`.log` the log: everything Stata printed while the do-file ran. The thing you hand in

`.ado` a command that somebody wrote in Stata. `help` shows it like any other; `ssc install` fetches new ones

the do-file makes the log from the data. Keep all three together in one folder

## . What you get

`wage_survey.dta` on Brightspace: the dataset, 8,000 simulated wage workers, made to look like a real Indonesian survey

`exercise.html` on Brightspace: the tasks we do together, with every answer folded away until you click. Open it now

`session.do` on Brightspace: the finished do-file, to check your own against afterwards

nothing to install. save the download in a folder you can find again

## . Ten minutes: tell Stata where your folder is

`cd "C:/Users/you/Downloads/intro_to_stata"` your folder, wherever you saved it. Find it once: File > Change Working Directory, pick it, then type `display c(pwd)` and copy the answer

`capture log close` closes a log if one is open, and says nothing if not

`log using "session.log", replace` from here on, everything is written down

`use "wage_survey.dta", clear` opens the data; `clear` drops whatever was in memory first

the first four lines of your do-file. Type them, run them.

“file not found” means the folder line. Raise a hand; that is what the ten minutes are for

## . The email arrives

Hi, I am writing the wage section and need five things from the survey by Thursday.

- 1-4 how many people do we actually have, and what do they look like?
- 5 what does a typical worker earn?
- 6 what share finished secondary school, and is it lower for people who grew up poor?
- 7-9 what is the return to a year of schooling, holding province, gender, age and location fixed?
- 10 does that return look different for people who grew up poor?

Send me the numbers and how you got them. Thanks!

eleven tasks, one command or two each. Run it, read it, then we talk

## . 1. Open the data

*the email asks: how many people do we actually have?*

`use "wage_survey.dta", clear` again, from the do-file: a do-file works from the top

run it. What did Stata print?

## . 1. Open the data

*the email asks: how many people do we actually have?*

`use "wage_survey.dta", clear` again, from the do-file: a do-file works from the top

run it. What did Stata print?

One line, the data label. Stata is quiet when things work.

## . 2. How many people, and what is in the file

*the email asks: how many people do we actually have?*

`describe` every variable, its label, how many rows

how many observations, how many variables?

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*the email asks: how many people do we actually have?*

`describe` every variable, its label, how many rows

how many observations, how many variables?

8,000 and 10. “Actually” is doing some work in that question.

### . 3. What do they look like

*the email asks: ...and what do they look like?*

`summarize` mean, min, max, N of every number

two numbers in that table should bother you. Which?

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*the email asks: ...and what do they look like?*

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two numbers in that table should bother you. Which?

The maximum age is 320, and schooling has 7,800 answers, not 8,000.

For the reply: 40 percent women, 55 percent urban, 28 percent grew up poor, mean age 25, mean schooling 10 years.

## . 4. But first, those ages

*the email asks: how many people do we actually have?*

`histogram age` one crushed bar: a picture shows a wrong value first

`list id age birthyr if age > 100` the people above 100, with year of birth

ten people. What happened to them, and what can you do about it?

## . 4. But first, those ages

*the email asks: how many people do we actually have?*

histogram age one crushed bar: a picture shows a wrong value first

list id age birthyr if age > 100 the people above 100, with year of birth

ten people. What happened to them, and what can you do about it?

An extra zero: born 1994, surveyed 2014, so 20, not 200. The file knows the right value, so repair it:

```
replace age = 2014 - birthyr if age > 100
```

Ten changes; ages run 15 to 35 again.

## . 5. What does a typical worker earn

*the email asks: what does a typical worker earn?*

`summarize hwage, detail` one variable, with percentiles

mean, median, 25th and 75th percentile. Which one answers the question?

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`summarize hwage, detail` one variable, with percentiles

mean, median, 25th and 75th percentile. Which one answers the question?

Mean 11,218, median 8,571. A few high earners pull the mean up. Report the median, and say so.

## . 6. Secondary school, by background

*the email asks: what share finished secondary school, and is it lower for people who grew up poor?*

```
generate high = schooling >= 12 a new variable: 1 if the condition holds, 0  
if not
```

```
tabulate high, missing count each value, and show the missing ones too
```

is the table right?

## . 6. Secondary school, by background

*the email asks: what share finished secondary school, and is it lower for people who grew up poor?*

```
generate high = schooling >= 12
```

 a new variable: 1 if the condition holds, 0 if not

```
tabulate high, missing
```

 count each value, and show the missing ones too

is the table right?

No. 5,220 zeros, 2,780 ones, no missing: the 200 without an answer all became 1. Stata stores missing as larger than any number.

## . 6, continued: say what you mean

drop high throw the wrong one away

generate high = schooling >= 12 if !missing(schooling) only for  
people who answered

tabulate high, missing 5,220 zeros, 2,580 ones, 200 missing. Right

tabulate cpoor high, row the cross-table, with row percentages

read the row percentages

## . 6, continued: say what you mean

drop high throw the wrong one away

generate high = schooling >= 12 if !missing(schooling) only for people who answered

tabulate high, missing 5,220 zeros, 2,580 ones, 200 missing. Right

tabulate cpoor high, row the cross-table, with row percentages

read the row percentages

38 percent of the never-poor, 21 percent of those who grew up poor.

every > or >= on a variable that can be missing needs & !missing(). In SAS missing is smaller than any number, so there the trap sits in <.

## . 7. The return to a year of schooling

*the email asks: the return to a year of schooling, holding province, gender, age and location fixed*

`generate lhwage = ln(hwage)` the log of the wage: a change of 0.01 is about 1 percent

`regress lhwage schooling female urban age i.province i.province`  
turns one variable into a dummy per province

read two numbers: N, and the coefficient on schooling

## . 7. The return to a year of schooling

*the email asks: the return to a year of schooling, holding province, gender, age and location fixed*

`generate lhwage = ln(hwage)` the log of the wage: a change of 0.01 is about 1 percent

`regress lhwage schooling female urban age i.province` `i.province` turns one variable into a dummy per province

read two numbers: N, and the coefficient on schooling

N 7,800: the 200 without schooling are silently left out. Schooling 0.079: a year of schooling is associated with an 8 percent higher wage.

Skipped task 4? Then look at the coefficient on age.

## . 8. Did the model behave

if time allows

`predict yhat` the fitted value, from the regression still in memory

`predict uhat, residuals` actual minus fitted

`summarize yhat uhat` their means, and their N

what is the mean of the residuals, and why?

## . 8. Did the model behave

if time allows

`predict yhat` the fitted value, from the regression still in memory

`predict uhat, residuals` actual minus fitted

`summarize yhat uhat` their means, and their N

what is the mean of the residuals, and why?

Zero, by construction: OLS picks the coefficients that make the residuals average zero. Both N are 7,800.

## . 9. One number for the relationship

if time allows

`correlate hwage schooling` +1 a perfect line up, -1 a perfect line down, 0  
none

what does this number tell you that the regression does not, and the reverse?

## . 9. One number for the relationship

if time allows

`correlate hwage schooling` +1 a perfect line up, -1 a perfect line down, 0 none

what does this number tell you that the regression does not, and the reverse?

0.34. One number, no units, nothing held fixed. The regression says by how much.

## . 10. Different for people who grew up poor?

*the email asks: does that return look different for people who grew up poor?*

The same regression as task 7, only for people who grew up poor. No command on this slide: build it from task 7. You know the variable, and you met `if` in task 4.

run it, then compare the coefficient on schooling with task 7

## . 10. Different for people who grew up poor?

*the email asks: does that return look different for people who grew up poor?*

The same regression as task 7, only for people who grew up poor. No command on this slide: build it from task 7. You know the variable, and you met `if` in task 4.

run it, then compare the coefficient on schooling with task 7

```
regress lhwage schooling female urban age i.province if cpoor == 1
```

0.060 against 0.079, on 2,166 people. A smaller gain for those who grew up poor. Causal? Another course.

## . 11. Close the log, write the reply

*the email asks: send me the numbers and how you got them*

`log close` then open `session.log` in a text editor

five lines: the number, and the command in your log that produced it

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*the email asks: send me the numbers and how you got them*

`log close` then open `session.log` in a text editor

five lines: the number, and the command in your log that produced it

|                                                           |                             |
|-----------------------------------------------------------|-----------------------------|
| 8,000 people, ten ages repaired, 200<br>without schooling | describe, summarize, task 4 |
| median wage 8,571                                         | summarize hwage, detail     |
| secondary school: 38% never poor,<br>21% grew up poor     | tabulate cpoor high, row    |
| return 0.079 per year, N 7,800                            | regress                     |
| 0.060 for the childhood poor, N 2,166                     | regress ... if cpoor == 1   |
| New data? Change one file name, run it again.             |                             |

## . Everything you typed today

set up

```
cd "..."
```

```
capture log close
```

```
log using "session.log"
```

```
use "wage_survey.dta"
```

where your folder is

close any open log

start the record

open the data

look

```
describe
```

```
summarize
```

```
summarize hwage, detail
```

```
tabulate cpoor high
```

```
browse
```

what is in the file

mean, min, max, N

with percentiles

count, cross-count

look at the data

fix

```
histogram age
```

```
list if age > 100
```

```
replace age = ...
```

```
generate high = ...
```

one picture

show some rows

change values

a new variable

estimate

```
regress lhwage schooling
```

```
predict yhat
```

```
correlate hwage schooling
```

```
log close
```

```
help regress
```

the regression

fitted values

one number

end the record

the manual

eighteen lines. Every one of them is in session.do on Brightspace, with the notes

## . When you are stuck

`help regress` Stata's own manual for any command. Try it now

`search missing values` when you do not know the command's name

`an AI assistant` good at reading an error message. Check any command it proposes with `help` first; if `help` has never heard of it, it is not real

paste the whole error, the command, and what you were trying to do

## . The second email

“Thanks, this is great. The new survey wave just came in. Could you redo it all by Thursday?”

five minutes, because you have a do-file

questions, any time:

`a.febriady@rug.nl`